

**Program: Computer Engineering**

**Instructors: Dr. Seham Mowad, TA: Silvana Karim**

**Date: 10 March 2026**

**Course/Code: Computer Organization and Architecture CSE221**

**Duration: 2 Hours**

**Q1:** What general categories of functions are specified by computer instructions?

**Q2:** List and briefly define two approaches to dealing with multiple interrupts.

**Q3:** List and briefly define the possible states that define an instruction execution.

**Q4:** What is the benefit of using a multiple-bus architecture compared to single-bus architecture?

**Q5:** The hypothetical machine of Figure 3.4 also has two I/O instructions:

0011 Load AC from I/O

0111 Store AC to I/O

In these cases, the 12-bit address identifies a particular I/O device.

Show the program execution (using the format of Figure 3.5) for the following program:

1. Load AC from device 5.
2. Add contents of memory location 940.
3. Store AC to device 6.

Assume that the next value retrieved from device 5 is 3 and that location 940 contains a value of 2.

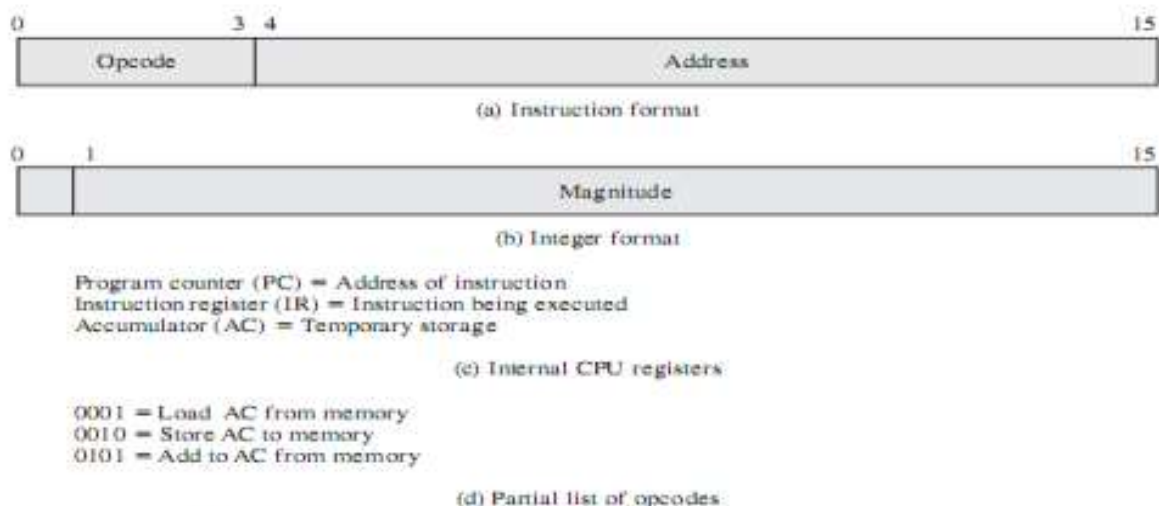


Figure 3.4 Characteristics of a Hypothetical Machine

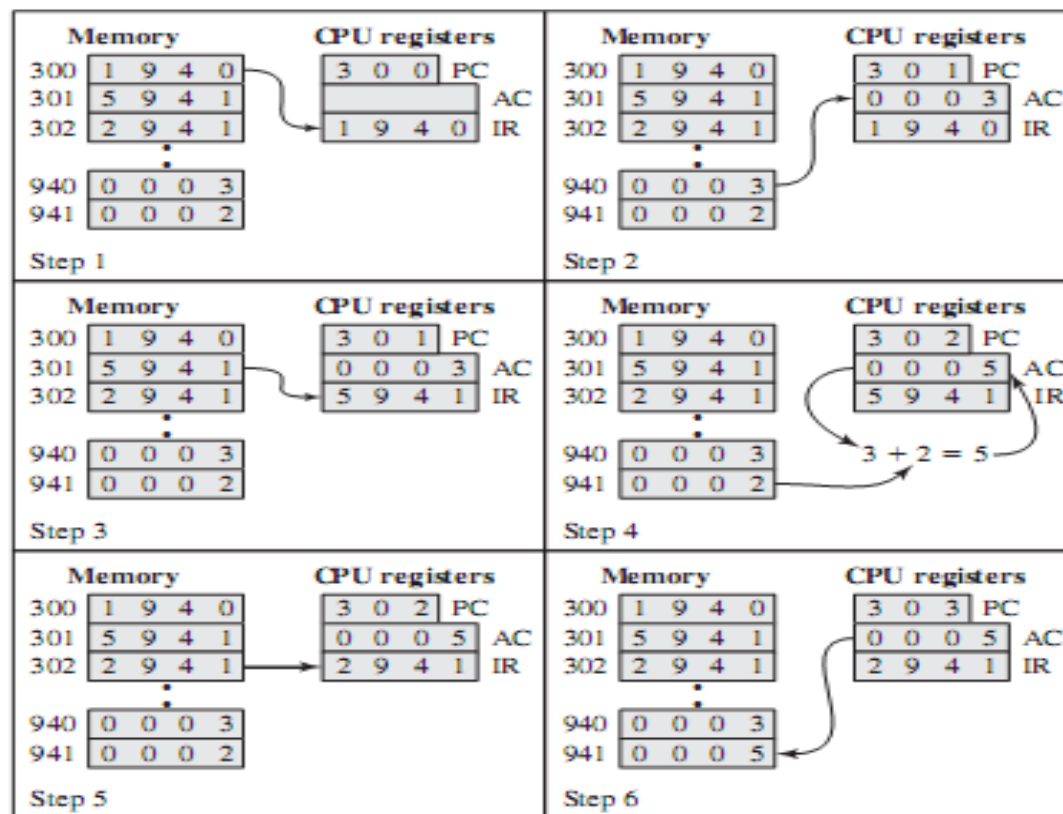


Figure 3.5 Example of Program Execution (contents of memory and registers in hexadecimal)

**Q6:** The program execution of Figure 3.5 (in the previous question) is described in the text using six steps. Expand this description to show the use of the MAR and MBR.

**Q7:** Consider a hypothetical 32-bit microprocessor having 32-bit instructions composed of two fields:

The first byte contains the opcode and the remainder the immediate operand or an operand address.

- What is the maximum directly addressable memory capacity (in bytes)?
- Discuss the impact on the system speed if the microprocessor bus has
  - a 32-bit local address bus and a 16-bit local data bus.
  - a 16-bit local address bus and a 16-bit local data bus.
- How many bits are needed for the program counter and the instruction register?

**Q8:** Consider a hypothetical microprocessor generating a 16-bit address (for example, assume that the program counter and the address registers are 16 bits wide) and have a 16-bit data bus.

- What is the maximum memory address space that the processor can access directly if it is connected to a “16-bit memory”?
- What is the maximum memory address space that the processor can access directly if it is connected to an “8-bit memory”?

c. What architectural features will allow this microprocessor to access a separate “I/O space”?

d. If an input and an output instruction can specify an 8-bit I/O port number, how many 8-bit I/O ports can the microprocessor support? How many 16-bit I/O ports? Explain.

**Q9:** Consider a 32-bit microprocessor, with a 16-bit external data bus, driven by an 8-MHz input clock. Assume that this microprocessor has a bus cycle whose minimum duration equals four input clock cycles. What is the maximum data transfer rate across the bus that this microprocessor can sustain, in bytes/s? To increase its performance, would it be better to make its external data bus 32 bits or to double the external clock frequency supplied to the microprocessor? State any other assumptions you make and explain.

**Hint:** Determine the number of bytes that can be transferred per bus cycle.

**Q10:** Consider a computer system that contains an I/O module controlling a simple keyboard/printer teletype.

The following registers are contained in the processor and connected directly to the system bus:

- INPR: Input Register, 8 bits
- OUTR: Output Register, 8 bits
- FGI: Input Flag, 1 bit
- FGO: Output Flag, 1 bit
- IEN: Interrupt Enable, 1 bit

Keystroke input from the teletype and printer output to the teletype are controlled by the I/O module. The teletype can encode an alphanumeric symbol to an 8-bit word and decode an 8-bit word into an alphanumeric symbol.

a) Describe how the processor, using the first four registers listed in this problem, can achieve I/O with the teletype.

b) Describe how the function can be performed more efficiently by also employing IEN.

**Q11:** Consider two microprocessors having 8- and 16-bit-wide external data buses, respectively. The two processors are identical otherwise and their bus cycles take just as long.

a) Suppose all instructions and operands are two bytes long. By what factor do the maximum data transfer rates differ?

b) Repeat assuming that half of the operands and instructions are one byte long.

**Q12:** Consider a microprocessor that has memory reading timing, after some analysis, a designer determines that the memory falls short of providing read data on time by about 180 ns.

**a)** How many wait states (clock cycles) need to be inserted for proper system operation if the bus clocking rate is 8 MHz?

**b)** To enforce the wait states, a Ready status line is employed. Once the processor has issued a Read command, it must wait until the Ready line is asserted before attempting to read data. At what time interval must we keep the Ready line low to force the processor to insert the required number of wait states?